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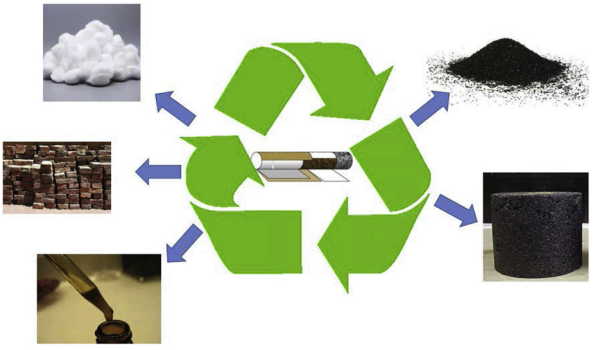
Review

A second life for cigarette butts? A review of recycling solutions

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GRAPHICAL ABSTRACT



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ABSTRACT

Trillions of cigarettes are smoked annually making cigarette butts one of the most common types of litter in the world. Due to the materials and toxic substances that they contain, this waste carries a very harmful risk for the environment and for living organisms (including humans). Only a few - barely sustainable - solutions have tried to tackle this waste and alternative solutions to landfilling and incineration are needed. Identifying the best methodological solutions and technologies for recycling this kind of waste in terms of results and applicability to real contexts would reduce the presence of dangerous materials in the environment and ecosystems and would promote the recovery of materials in line with the circular economy and sustainable development.

The objective of this review was to collect and analyze the alternative solutions available in the literature for the recovery and recycling of the materials in cigarette butts, considering them as possible sources of secondary raw materials applicable to contexts of common interest. Several papers were selected and the results obtained by the authors are presented in terms of type of treatment process (physical, chemical or both), product derived (in solid, liquid or gaseous form) and its possible use in different sectors (e.g. construction, electronics, energy, chemistry and environmental protection). The main results, together with the advantages and disadvantages are highlighted and proposals for further research are outlined.

Abbreviations: CF, cigarette filter "component of a cigarette specifically designed to absorb vapors and to accumulate particulate smoke components". In this paper, the authors use CF as a clean component before its use during cigarette smoking; CB, cigarette butt "part of a cigarette that is left after it has been smoked"; CA, cellulose acetate "acetate ester of cellulose, also used in the manufacture of cigarette filters"

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1. Introduction

The world faces numerous environmental challenges on a daily basis. Efforts to address them are aimed at facilitating sustainable and responsible growth.

As described by the World Health Organization (WHO, 2017), “Tobacco threatens many of the Earth’s resources”. The WHO study analyses the life cycle impacts of tobacco and shows how this product (and its industry) creates waste and inflicts damage on the environment throughout its life cycle. It also examines how this requires all those involved with tobacco control to think about how these consequences are linked to sustainable development goals.

This is a global challenge because, although the cultivation, harvesting and processing of tobacco are limited to specific areas of the world, the consumption and post-consumer waste are a problem that affects the entire planet (inhabited and uninhabited places). This is waste that can be found everywhere, which is often difficult to collect and even more complex to treat properly. Furthermore, the corresponding costs are not necessarily borne by the manufacturers, distributors, or users of tobacco products, but generally are the responsibility of all citizens.

Worldwide, an estimated 967 million daily smokers consumed approximately 6.25 trillion cigarettes over the course of 2012 (Ng et al., 2014). According to the Tobacco Atlas (https://tobaccoatlas.org), a project of the American Cancer Society and the Vital Strategies, in 2016 the number of cigarettes smoked decreased to 5.7 trillion. Despite this, the number is expected to reach nine trillion by 2025 due to further growth in the market (Mackay et al., 2006). This is mainly due to population growth (Srbinoska et al., 2018). The number of cigarettes per smoker per day also varies widely across countries (Ng et al., 2014).

Fig. 1 shows the global cigarette consumption per region between 1999 and 2016. The trend highlights the regions affected by higher growth (China and South-East Asia Region - SEARO) and those characterized by a decline (European Region -EURO, Americas - AMRO, Western Pacific Region - WPRO - excluding China). This trend is reflected in the per capita consumption and on the waste associated with cigarette consumption, as shown in Figs. 2 and 3.

CBs are the main type of litter: they are one of the most frequently collected waste items during clean-ups, in particular in beach and coastal environments (Araújoa and Costa, 2019; Ocean Conservancy, 2018). The estimated annually discarded waste from global cigarette consumption could be anywhere between 340–680 million kg, besides almost two million tonnes of paper, ink, cellophane, foil and glue that are used in tobacco product packaging. More than 40% of all items collected during urban clean-ups and removed from public areas such as beaches and parks are CBs and related products (Bonanomi et al., 2015; Ariza et al., 2008). According to WHO (2017), up to two-thirds of every smoked cigarette is discarded on the ground, and about 76% according to Patel et al. (2013). This situation is explained by Patel et al. (2013) and Novotny and Zhao (1999): through interviews, they found that 73.5% of smokers do not use correct practices for the disposal of CBs, which are often therefore released into the environment.

According to Roder Green et al. (2014), CBs are not uniformly distributed in an urban area, but are strongly connected to the location of sales and consumption. Their research (concentrated in the US) showed evidence of a higher concentration in proximity to bars, convenience stores, liquor stores, cafes, gas stations, grocery stores, restaurants, and traffic signals, as also reported by Marah and Novotny (2011) and Van Gucht et al. (2010). Furthermore, the authors report how CBs are concentrated at places of transition from an outdoor to an indoor environment (entrances to buildings, vehicles or public transportation stations).

The big problem associated with this waste, in addition to its lack of control as it is freely dispersed in the environment, is that it is not biodegradable and that it contains and releases over 7000 toxic chemicals in the environment.

Its toxicity and non-biodegradability could justify environmental regulation and policies to raise the price of tobacco and further discourage its use (Smith and McDaniel, 2010).

However, practical operational aspects are lacking, in particular at the regulatory level. For example, the European List of Waste (EC, 2000; EU, 2014) does not propose a specific entry for CBs, which is therefore considered as “Municipal waste including separately collected fraction/Separately collected fraction/Other fractions not otherwise specified” (Rebischung et al., 2018).

Therefore, using a "precautionary principle" and appropriate awareness raising and accountability practices among consumers and producers would substantially contribute to reducing the production of waste and its dispersion into the environment.

Currently, given the poor results, finding effective management solutions for this waste is essential as it has been widely demonstrated how conventional disposal methods mainly represented by landfilling or incineration are neither universally sustainable nor economically feasible for this purpose (Bandi et al., 2018; Mohajerani et al., 2016). Identifying a recycling methodology for this kind of waste would reduce the release of dangerous materials into the environment and to the ecosystems, and would promote the recovery of materials in line with the circular economy and sustainable development.

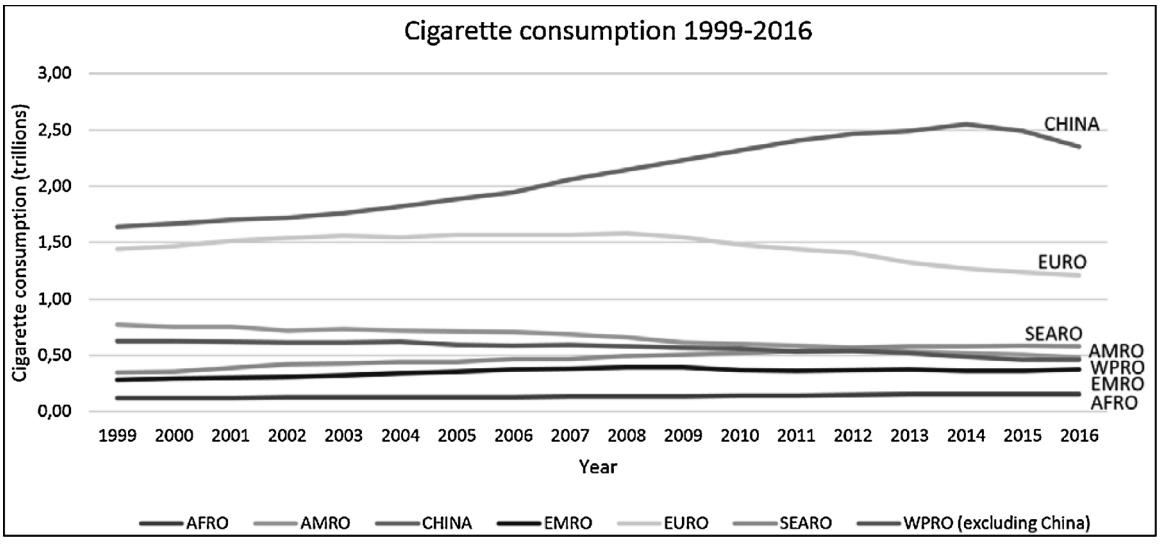


Fig. 1. Trend of global cigarette consumption per region (authors elaboration from Tobacco Atlas data).

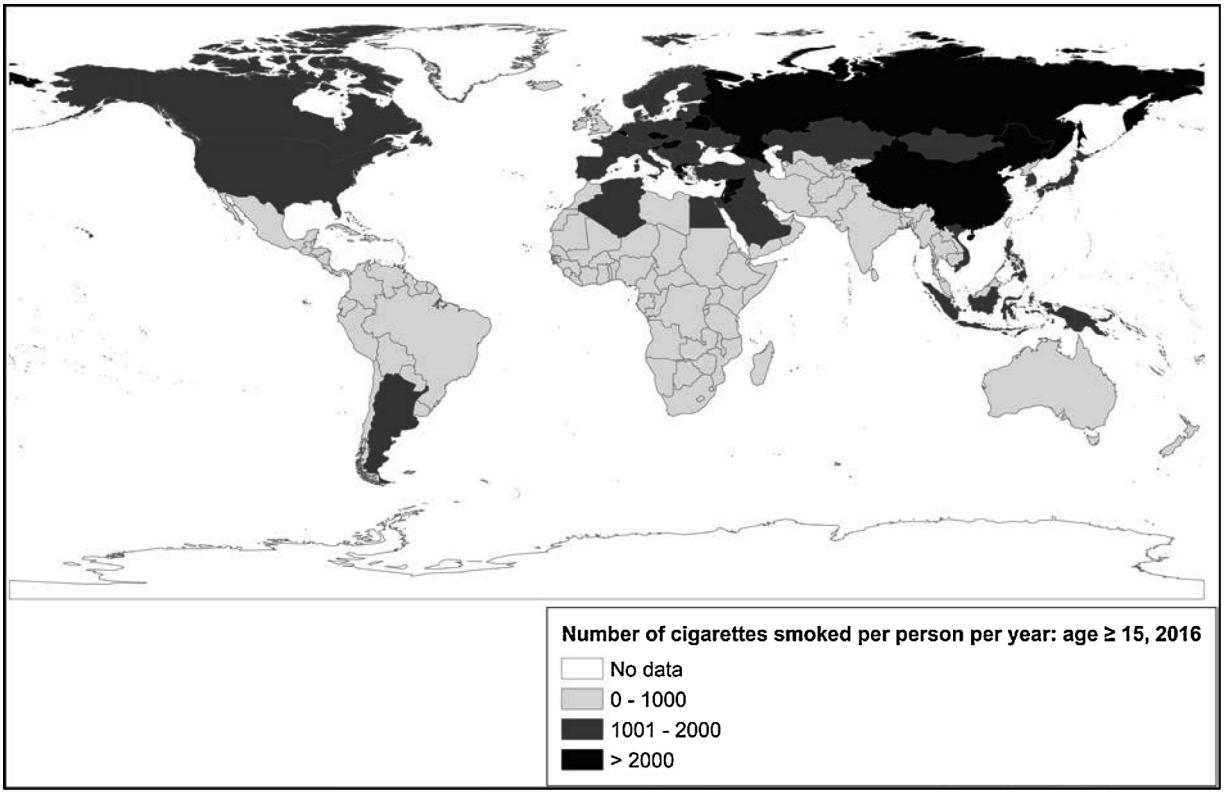


Fig. 2. Pro-capita cigarette consumption (authors elaboration from Tobacco Atlas data).

Several studies have analysed the possible treatments of CBs to make them less harmful and to give the waste a second life through the production of secondary raw materials or products that are already usable on the market. With the concept of reverse logistics and considering that the CBs are mainly made up of cellulose acetate (CA), these studies analyse their value as potential cellulose derivative sources for industrial and chemical processes. Torkashvand and Farzadkia (2019) also reviewed various solutions to cigarette butt

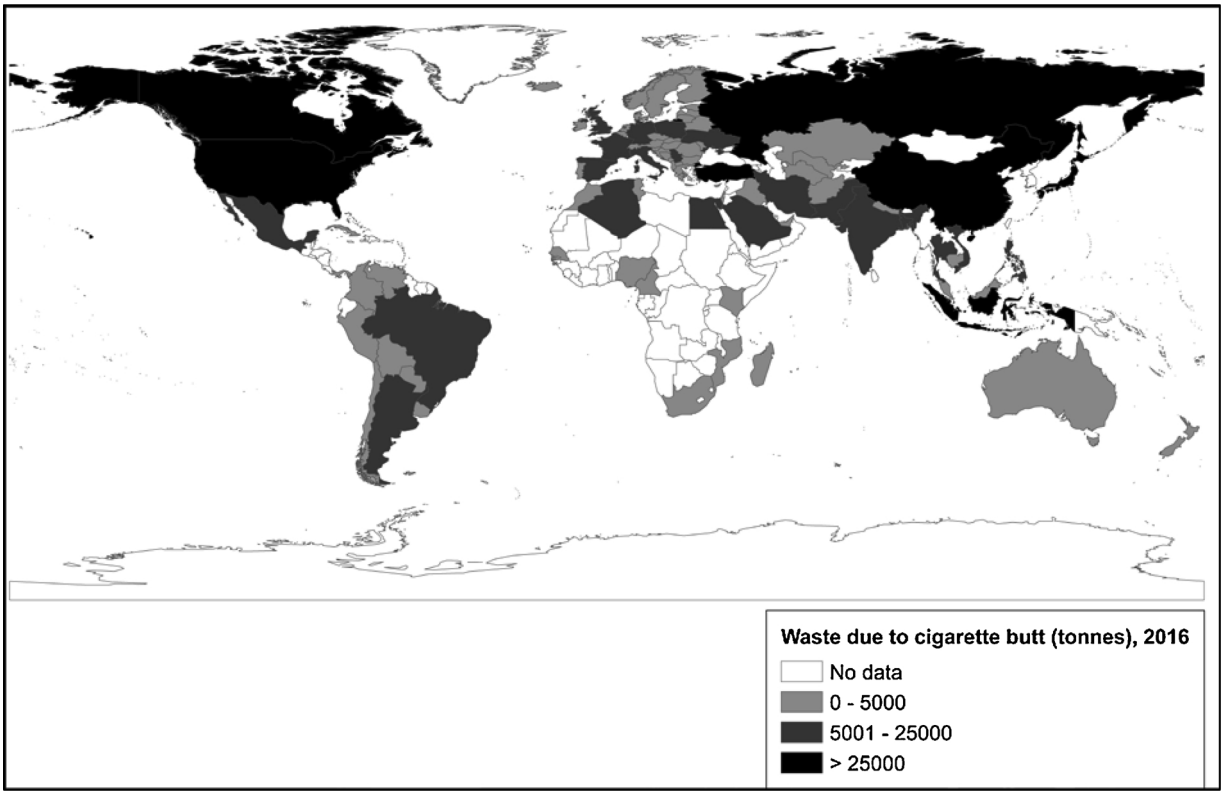


Fig. 3. Waste due to cigarette butt (authors elaboration from Tobacco Atlas data).

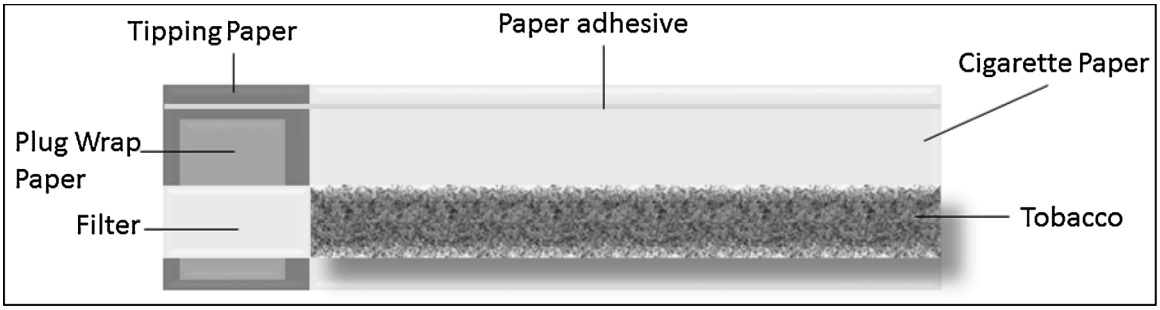


Fig. 4. Diagram of a cigarette (authors elaboration).

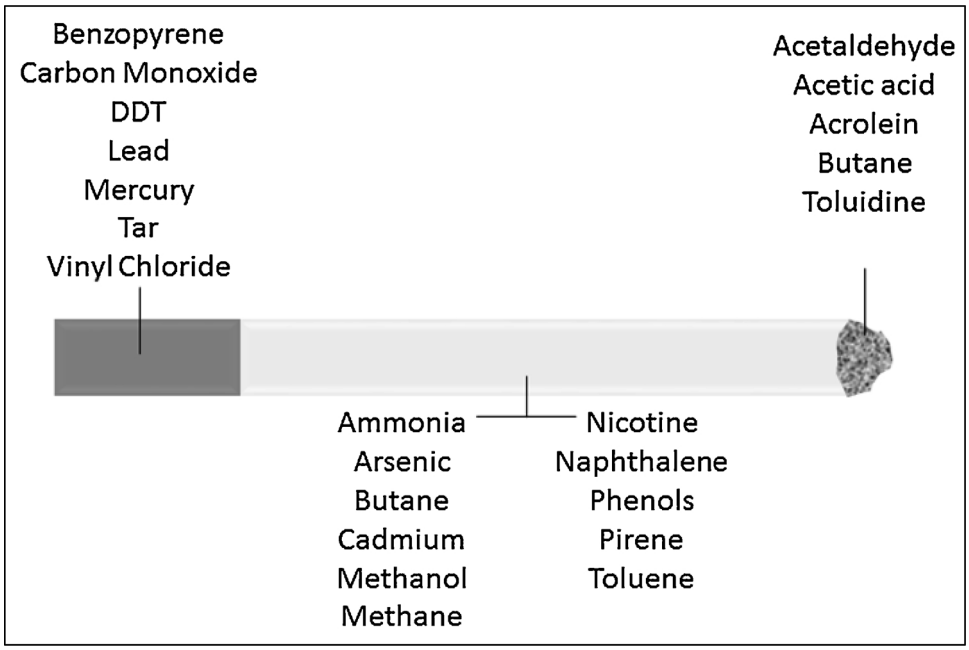


Fig. 5. Composition of Cigarettes (authors elaboration).

management.

The present work presents a critical review of the research available in the literature concerning the recycling of CBs, as an inert component of other high value-added materials. Particular attention is paid to the evaluation of treatment approaches and related derived products. The geographical distribution of the analysed studies is also assessed. These aspects provide additional insights to those described by Torkashvand and Farzadkia (2019).

These results can be of great interest to territories around the world and of inspiration to apply solutions and technologies to eliminate waste that has a huge environmental impact and that can help to develop other products according to the circular economy paradigm, avoiding the use of raw materials and precious natural resources.

2. Characteristics of cigarette filters and CBs

A cigarette is a small cylinder of finely cut tobacco leaves rolled in thin paper.

Modern commercially manufactured cigarettes are composed of the following main elements (Fig. 4):

- Tobacco blend dried and cut into thin one millimeter thick strips. The characteristics of tobacco change from manufacturer to manufacturer. Dangerous chemicals can form in and be deposited on tobacco during the processing of the tobacco leaves. Other dangerous chemicals are created when the tobacco filler is burned. Often there are also various additives that manufacturers can add to make smoking more appealing and to mask the harshness of the smoke. Humectants such as propylene glycol or glycerol, as well as flavouring products and enhancers such as cocoa solids, liquorice, tobacco extracts, and various sugars are some of the additives in cigarettes. There are hundreds of possible additives, however these are not declared on the cigarette packages on the market.
- Rolling paper to cover the tobacco and adhesive. Composed of cellulose (with possible additives), this paper favours the appearance of the ash and guarantees a uniform combustion. It is closed by an adhesive substance.
- A filter made up of cellulose acetate, described in more detail below.
- Tipping paper to cover the filter. This is a polyvinyl acetate emulsion which is used to glue the cellulose acetate plug to the inner wrapping paper.

According to the U.S. Department of Health and Human Services (USDHHS), smokers are exposed to a toxic mix of over 7000 chemicals when they inhale cigarette smoke (U.S. Department of Health and Human Services (USDHHS, 2010). Hundreds of these substances are toxic, about 70 are carcinogenic.

In 2012 the U.S. Food and Drug Administration (FDA) published a preliminary list of 93 Harmful and Potentially Harmful Constituents (HPHCs) that are linked to the five most serious health effects of tobacco use (cancer, cardiovascular disease, respiratory effects, reproductive problems, and addiction). Some of these substances are

summarized in Fig. 5.

An important element that makes up the cigarette is the filter (CF). CFs were introduced in the middle of the 20th century, based on a patent by Boris Aivaz recorded in 1925.

A filter is designed to reduce the amount of smoke, tar and fine particles inhaled during the combustion and to reduce the harshness of the smoke and keep tobacco flakes out of the smoker's mouth. The filter captures and retains toxic substances, but when littered, such chemicals can leach out of the cigarette filter.

According to Mohajerani et al. (2016), almost 90% of all cigarettes are manufactured with cellulose acetate filter tips except in Japan, Venezuela, South Korea and Hungary.

CFs are made of about 95% poorly degradable microscopic-sized white fibres massed together (about 12,000 per filter with about 30,000 dernier), characterized by a Y-shaped cross-section that is not perpendicular to the flow (the angle ranges from 18° to 49°) (Du et al., 2015) and it is made of CA. These fibres are synthetic plastic which looks like cotton. In the CFs, the CA fibres are linked to each other through the glycerol triacetate, which is a plasticizer (Hamzah and Umar, 2017). CA is one of the most important esters of cellulose. It is produced through a two-step process, involving the esterification of acetic acid with cellulose and the partial hydrolysis of the resulting ester groups. CA can be produced with varying degrees of substitution. The most common form of cellulose acetate fibre has an acetate group on approximately 2–2.5 of every three hydroxyls (Fischer et al., 2008), due to good solubility in common solvents, molecular weights, and melt properties (Puls et al., 2011). CA has various applications which depend on its processing and its physico-chemical characteristics. In particular, it is used for the production of textiles, plastics, films, and CFs.

The common name for the remains of a cigarette after smoking is a CB. According to Poppendieck et al. (2016), a CB can be defined as “the cigarette remaining at the conclusion of the smouldering phase following smoking. The conclusion of the smouldering phase is defined as when the entire cigarette butt reaches the ambient temperature of the environment in which it is located. The CB includes any remaining paper (wrapping paper, tipping paper, and plug wrap paper)”. It is typically about 30% of the cigarette's original length.

The three major CB components that Poppendieck et al. (2016) listed are: 1) ash, 2) unburned tobacco and 3) filtration material (Fig. 6).

A filter, after its use, has absorbed smoke, trap tar, toxic chemicals and particle smoke. Its chemical composition depends on the type of tobacco used, the cigarette paper, category and effectiveness of the filter as well as the degree of tip ventilation. It also depends on the filter design. Strain et al. (2015) analyzed the smoke filtering properties of different fibres using a cigarette as the smoke source. The tests showed the presence of many carcinogenic hydrocarbons and alcohols that exist in tobacco smoke. Experiments have also shown that the original CA cigarette filter tip absorbed only 2.7 times its own weight during the same treatment.

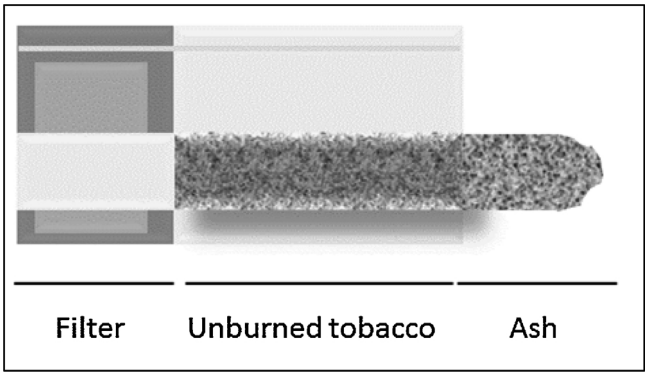


Fig. 6. Composition of a CB (authors elaboration).

The main problem associated with CBs is their slow degradation rate and the high concentration of toxic content.

Typically, cellulose acetate possesses poor photo- and bio- degradable properties, making CBs persistent in the environment. The methods and times required for this process depend on the degradation mechanism, of which there are different types, e.g. photodegradation, biodegradation, and mechanical processes or a combination of these as typically occurs in outdoor environments due to different factors. Some researchers have demonstrated how the speed of degradation processes are inversely proportional to time. In particular, Puls et al. (2011) carried out a review of degradation mechanisms and the potential degradation rates of CA. According to Puls et al. (2011), cellulose is degradable thanks to organisms that utilize cellulose enzymes. CA, due to the additional acetyl groups, requires the presence of esterases for the first step in degradation. Also, the high molecular weight, crystallinity and the physical form influence the microbial degradation efficacy (Haske-Cornelius et al., 2017; Ogundare et al., 2017). Furthermore, CA is photo-chemically degraded by UV, but its property is limited to photo degradability in sunlight due to the lack of chromophores for absorbing ultraviolet light (Robertson et al., 2012). Thus cellulose acetate-based cigarette filters do not biodegrade under most circumstances because of their compressed make up and the presence of acetyl molecules (Haske-Cornelius et al., 2017).

CBs require several years to degrade completely (Bonanomi et al., 2015; Puls et al., 2011). According to Kadir et al. (2009), the degrading rates vary from within 1 to 2 months under anaerobic conditions, 6 to 9 months in soil, 12 months in fresh water and 36 months or more under poor conditions, for example, sea water, desert and underground stations. Bonanomi et al. (2015) has demonstrated this slow process of degradation through a 2-year experiment in a standard decomposing organic substrate. The study shows how CBs lost only 37.8% of their initial mass after two years of decomposition. Comparing cellulose and plastic cigarette filter decomposition, Joly and Coulis (2018) estimated that conventional plastic filters take 7.5–14 years to disappear in compost and on the soil surface, respectively. In contrast, they estimated that cellulose filters take 2.3–13 years to disappear, in compost and on the soil surface, respectively.

The other significant problem is the over 7000 toxic chemicals (including up to 50 human carcinogens according to Hoffmann and Hoffmann (1997), toxic heavy metals, polycyclic aromatic hydrocarbons, pesticides, N-nitrosamines, aromatic amines, formaldehyde, acetaldehyde, benzene and nicotine) that this waste releases into the water, the soil, and the air, and which then comes into direct contact with humans, animals and vegetation.

Poppendieck et al. (2016) provided a very in-depth review of the chemicals in cigarette butt emission and extraction studies, divided by media (air, water, extract solution, cigarette filter, cigarette butt), which summarized the chemicals found in the environment are a potential threaten to human health and the environment, especially marine ecosystems. These problems, in particular associated with harmful chemicals, have been analysed by several other authors.

Roder Green et al. (2014) studied the effects of the nicotine contained in CBs on urban water resources through batch and rainfall experiments. Slaughter et al. (2011) analysed the effects on fish exposed to CBs soaked in either fresh or salt water. Singh and Kathiresan (2015) analysed the effects on fish and microbial communities. Oropesa et al. (2017) studied the effects on aquatic organisms from different trophic levels. This process revealed that nicotine poses an important risk to aquatic organisms. Booth et al. (2015) tested how common intertidal molluscs were affected by butt leachate.

Slaughter et al. (2011) showed that percolates from CBs are toxic to a range of marine organisms. The LC50 (lethal concentration capable of killing 50% of the animals used in the test) is 1 CB/l for both marine and freshwater species.

Starkey (2018); de Granda-Orive et al. (2016); Novotny and Slaughter (2014); Bonanomi et al. (2015) and Kadir and Mohajerani

Table 1
Summary of literature searches.

Search keywords	No. of papers	Search date
Cigarette Butt	423 (1) 2,579 (2)	1 st January 2019
Cigarette Filter	1,867 (1) 19,869 (2)	1 st January 2019
Cigarette Tip	265 (1) 10,440 (2)	1 st January 2019
Cigarette Butt*recycling	14 (1) 244 (2)	1 st January 2019
Cigarette Filter*recycling	12 (1) 1,375 (2)	1 st January 2019
Cigarette Tip*recycling	0 (1) 619 (2)	1 st January 2019

(1)Scopus.
(2)ScienceDirect.

(2008) provided a review of the studies on living organisms and ecosystem health.

3. Methodology

This paper reviews the recent studies on CB recycling solutions. The scientific literature contains many interesting examples; however, the present review focuses only on specific experiences of CBs, which used this waste as input for physical and/or chemical treatment in order to obtain second raw materials, semi-finished products and finished goods applicable in industrial, agricultural, civil engineering, environmental or chemical contexts.

The research was conducted on papers published prior to 1 January 2019 using ScienceDirect and Scopus databases. The keywords used were: “Cigarette Butt”, “Cigarette Filter” and “Cigarette Tip”. These keywords were searched for individually or coupled with the term "recycling" using the "*" wildcard character used for multi-word searches. The results of the search, expressed as the number of papers identified, are summarized in Table 1.

Despite the numerous papers available, most were excluded from this work as they were not related to the topic discussed in this document. In fact, CBs are the subject of numerous related studies, for example, assessing environmental impacts and ecotoxicity, pollutant emission rates, physical dispersion, health effects, etc.

The papers identified were further filtered by selecting only those conforming to the focus of the present study: articles that directly studied possible ways for the recovery and recycling of CBs.

A total of 37 papers were thus selected which consider the definition of CBs, their chemical composition, techniques to treat and purify the smoked CBs, the products and materials that can be obtained and the performances of the products obtained.

The papers selected were classified using criteria that describe the process input, sample treatment system, output and possible use. The geographical distribution of the studies was also considered. The five criteria used were (Fig. 7):

1 Input for recycling: identifies the types of samples used as input to the treatment process. CBs are differentiated according to the filter production and characteristics of the smoked tobacco. Typically, this type of study compares CBs smoked with new and unused CFs (Table 2). Few authors used filters smoked in the laboratory to mimic the smoking process.
2 Methodology/Treatment system: identifies the type of approach adopted for the treatment of samples, identifying methods of chemical treatment, physical treatment and physical-chemical treatment. A brief description is also provided aimed at highlighting the methodological steps used by the authors (Table 3).
3 Type of product obtained from the process: characterizes the type of product obtained from the treatment process, identifying its physical state be it a solid, liquid or gaseous product. This criterion characterizes the output of the treatment process (Table 4).
4 Possible uses: describes the results of the process and describes its possible use. This criterion characterizes the possible use of the output obtained (Table 5).
5 Geographical distribution: identifies the country where the study was conducted (Table 6).

The results obtained from the evaluation of each criterion are described in the Section 4.

4. Results

Considering the final objective of the present study and assessing the results of recent studies on CB recycling solutions, the outcomes of the review are presented by individually analysing the criteria identified and described in Section 4.

4.1. Sample input

The reviewed papers are classified in Table 2 according to the type of CBs used as input to the treatment and recycling process.

The analysed papers apply approaches and techniques using smoked or unsmoked filters. The use of one or the other, or a comparison of the performance of the type of input used, depends on the authors’ methodological approach.

Of the 37 selected contributions, most of the authors (about 97%) used smoked CBs to develop and test recycling solutions. Only two cases (about 3%) used mimicked smoked CBs (one author) or supplementary materials (one author) as a precursor to the treatment and recovery process. Only five papers (about 14%) proposed a comparison using unused CFs. This is important because it checks whether and how the recycling process could influence the results, comparing the use of filters to "pure" material (cellulose acetate).

This comparison was conducted by Blankenship and Mokaya (2017), who used cigarette filters / butts to form the precursor for porous carbons in order to evaluate their hydrogen storage capacity. The results showed that a sequential benign hydrothermal carbonisation and activation treatment obtained a super-porous and ultra-high surface area ($4300\ m^2\ g^{-1}$) and pore volume ($2.09\ cm^3\ g^{-1}$) arising almost entirely (> 90%) from micropores. These characteristics, combined with the oxygen-rich nature of the material, demonstrated the highest hydrogen storage capacity reported in all the papers analysed. The comparison between smoked and unused filters highlighted various anomalies of the smoked samples mainly due to the presence of heavy metals.

Polarz et al. (2002) studied the transformation of cellulose acetate filters into micro-meter or nano-meter porous carbon structures. Applying an easy direct process, filters from smoked or unsmoked cigarettes were transformed into non-graphitic carbon (defined by the International Union of Pure and Applied Chemistry as “*all varieties of solids consisting mainly of the element carbon with two-dimensional long-range order of the carbon atoms in planar hexagonal networks, but without any measurable crystallographic order in the third direction (c-direction) apart from more or less parallel stacking*” and with many concrete applications such as heat shields, Li-ion batteries, activated carbons). The resulting material exhibited a well-defined mesoporosity, with different characteristics from the cellulose acetate fibres used as a precursor and with a good potential application in the field of gas purification or more generally as filters.

Gómez Escobar and Maderuelo Sanz (2017) analyzed the performance of smoked and unused CBs as absorbing materials. Each sample was obtained by collecting 10 butts inside an impedance tube. The samples differed in the type of butt used and their physical characteristics (length and diameter). All showed a high sound absorption

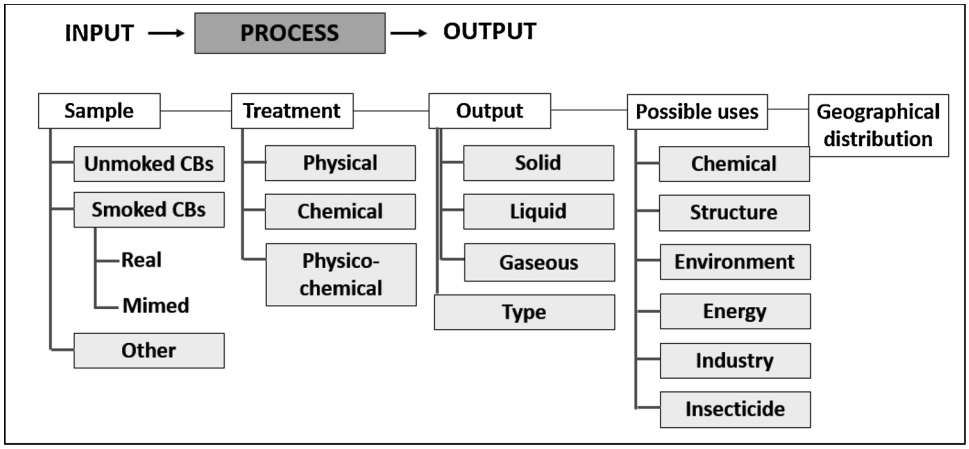


Fig. 7. Classification of features.

performance which, compared to some absorption materials (foam and glass wool), also exhibited great potential for commercial purposes. Maderuelo-Sanz et al. (2018) also presented a further development of their previous study, investigating only the acoustic performance of the cellulose acetate and not of the intact cigarette butts. The acoustic performance of used and not used CFs is similar (with a slightly better performance in terms of low frequencies for unused filters).

Murugan et al. (2018) investigated the use of CBs as a precursor for the preparation of mosquito pesticides and as tools for malaria vector management. The authors proposed a novel approach for the synthesis of Ag nanostructures that are toxic to young instars of the malaria vector Anopheles stephensi, chloroquine (CQ)-resistant malaria parasites, Plasmodium falciparum and microbial pathogens. Freshly smoked cigarette butts (i.e., < 3 h after smoking) with tobacco, without tobacco and unsmoked cigarette butts were used to produce CB extracts tested as pesticides. No mortality was observed by applying the extract from unsmoked cigarette butts, whereas low doses of CB extracts showed larvicidal and pupicidal toxicity, but without decreasing non-target risks.

Dieng et al. (2013) also conducted a study using CBs on insecticide-

Table 2
Type of CBs used for the recycling process.

	Smoked CBs	Mimicked CBs smoked	Unused CFs	Other
Anmei et al., 2018	X			
Bandi et al., 2018	X			
Cui et al., 2018	X			
Ifelebuegu et al., 2018	X			
Lima et al., 2018	X			
Maderuelo-Sanz et al., 2018	X		X	
Murugan et al., 2018	X		X	
Wadalkar et al., 2018	X			
Xiong et al., 2018a	X			
Xiong et al., 2018b	X			
Blankenship and Mokaya, 2017	X		X	
Ghosh et al., 2017	X			
Gómez Escobar and Maderuelo Sanz, 2017	X		X	
Hamzah and Umar, 2017	X			
Mohajerani et al., 2017	X			
Ogundare et al., 2017	X			
Suárez-Rodríguez and Garcia, 2017	X			
Teixeira et al., 2017	X			X Waste from CF production
Mohajerani et al., 2016	X			
Nasution, 2016	X			
Kim et al., 2016	X			
Wang et al., 2016	X			
Kadir and Mohajerani, 2015	X			
Lee et al., 2014	X			
Mondal et al., 2014	X			
Soltani et al., 2014	X			
Vahidhabanu et al., 2014	X			
Dieng et al., 2013		X		
Sarani and Kadir, 2013	X			
Suárez-Rodríguez et al., 2013	X			
Soltani and Yazdi, 2012	X			
Kadir and Mohajerani, 2011	X			
Sabzali et al., 2011	X			
Zhao et al., 2010	X			
Kadir et al., 2009	X			
Kadir and Mohajerani, 2008	X			
Polarz et al., 2002	X		X	

Table 3
Process description.

	Physical treatment	Chemical treatment	Physico- chemical treatment	Description
Anmei et al., 2018			X	Hydrothermal method: CBs were washed with water several times and then dried in the oven at 60 °C
Bandi et al., 2018			X	Hydrothermal method for the production of N, S co-doped carbon dots
Cui et al., 2018	X			Preparation of the bitumen mixture using the addition of CBs in the ordinary asphalt mixture
Ifelebuegu et al., 2018			X	Simple two-step: - cleaning and hydrolysis; - surface-functionalized
Lima et al., 2018			X	Low-temperature hydrothermal carbonization without using inert gas
Maderuelo-Sanz et al., 2018	X			CFs were dried, manually separated from the outer and inner paper and shredded into short fibres
Murugan et al., 2018		X		Hydrothermal process
Wadalkar et al., 2018	X			Incorporation of CBs into precast concrete blocks for paving
Xiong et al., 2018a			X	Preparation of the modified CF via SiO2, octadecyltrichlorosilane and methyltrimethoxysilane approach
Xiong et al., 2018b		X		Treatment steps: immersion in a pyrrole solution; oxidative polymerization with FeCl3; carbonization under nitrogen
Blankenship and Mokaya, 2017			X	Sequential benign hydrothermal carbonisation and activation
Ghosh et al., 2017		X		Filters removal and pyrolysis process
Gómez Escobar and Maderuelo Sanz, 2017				No treatment, just grouping the CBs
Hamzah and Umar, 2017			X	KOH chemical activation using a microwave input
Mohajerani et al., 2017	X			CBs encapsulated with different classes of bitumen and paraffin wax
Ogundare et al., 2017		X		Ethanolic extraction, hypochlorite bleaching, alkaline deacetylation, and conversion into NCC by sulfuric acid hydrolysis
Suárez-Rodríguez and Garcia, 2017				No treatment, just adding or removing CBs from nests
Teixeira et al., 2017			X	Separating all the cellulose content in the cigarette butts using four different processes: dissolving the ash and all chemicals adsorbed; hydrolyzing cellulose acetate into cellulose; removing lignin from the remaining tobacco and liberating its cellulose; disaggregation of the remaining paper
Mohajerani et al., 2016	X			CBs were disinfected at 105 °C and mixed in different percentages with brown silty clayey sand, dried, and fired
Nasution, 2016		X		CBs soaked in water for 10-15 days. The solution from CBW was then filtered
Kim et al., 2016		X		Heat treatments of the separated filters at 900 °C under NH3 and Ar, respectively. NCF-supported Pt catalysts were prepared by the conventional impregnation method
Wang et al., 2016		X		CBs soaked at 60 °C under magnetic stirring with NH4VO3 in distilled water
Kadir and Mohajerani, 2015	X			CBs disinfected and mixed in different percentages with brown silty clayey sand, dried, and fired
Lee et al., 2014			X	Heat treatment under a nitrogen-containing atmosphere
Mondal et al., 2014		X		CBs suspended in 50 ml distilled water
Soltani et al., 2014		X		Pyrolysis at 900 ∘C under a constant nitrogen gas flow in a steel tube furnace
Vahidhabanu et al., 2014		X		Isolate crude extracts from littered CBs using polar solvents
Dieng et al., 2013		X		CBs soaked in deionized water
Sarani and Kadir, 2013	X			CBs disinfected and mixed in different percentages with brown silty clayey sand, dried, and fired
Suárez-Rodríguez et al., 2013				No treatment, just adding or removing CBs from nests
Soltani and Yazdi, 2012			X	CBs mixed with 2 g powdered KOH
Kadir and Mohajerani, 2011	X			CBs disinfected and mixed in different percentages with brown silty clayey sand, dried, and fired
Sabzali et al., 2011			X	Preparation phase includes: separating tobacco from the cigarette; cutting the cigarette filters; soaking in water; separating filter paper wrap
Zhao et al., 2010		X		CBs were extracted in HCl solutions at 90 °C
Kadir et al., 2009	X			CBs disinfected and mixed in different percentages with brown silty clayey sand, dried, and fired
Kadir and Mohajerani, 2008	X			CBs disinfected and mixed in different percentages with brown silty clayey sand, dried, and fired
Polarz et al., 2002			X	CBs was transferred to a tube oven (Heraeus Thermicon P) equipped with a quartz tube. The tube was flooded with nitrogen prior to slow heating (1 °C/min) to a final temperature of 800 °C

Table 4
Results of the CBs treatment.

	Type of results	Physical state (S = solid; L = liquid; G = gas)
Anmei et al., 2018	Carbon Quantum Dots (CQDs) which are a recent type of carbon nanomaterial that exhibits strong fluorescence depending on the size and excitation wavelength. They are superior in terms of water solubility, photostability, toxicity and biocompatibility compared to traditional semiconductor QDs. CQDs are used in the chemical field to exploit their fluorescence in chemical analyses and, in particular, to detect synthetic colorants in foods.	S
Bandi et al., 2018	Carbon Dots (CDs) with a high fluorescence property. They exhibit unique optical properties, multiple functional groups, excellent biocompatibility and chemical and photo-stability. They have different practical applications: sensing, bioimaging, optoelectronic conversion, visible-light-activated bactericide, fingerprint detection, dye degradation, solar cells, printing inks and gene delivery. The authors describe the formation of CDs doping with non-metals such as N, S, B, and P. This allows to improve some of its properties, such as resistance to self quenching, enhanced fluorescence QY, and sensing selectivity.	S
Cui et al., 2018	Bituminous bitumen mixture, a composite material, formed from a mineral aggregate, a binder and asphalt which is laid in layers on the surface and compacted. In the case described by the authors, CBs have also been incorporated into the mixture, improving the strength, the anti-deformation and the high temperature resistance of the pavement.	S
Ifelebuegu et al., 2018	Oil-absorbent material, which can be used to mitigate the effects of possible oil spills in aquatic environments. This material guarantees low costs, easy and high efficiency operations, depending on the type of absorbent material used (natural, inorganic mineral or synthetic products). The material described by the authors, obtained from CBs, has demonstrated high absorption capacity and a good oil-saving rate, as well as being fabricated in a green and easy manner.	S
Lima et al., 2018	Hydrochar, a carbonaceous solid classifiable as lignite and with some possible market applications: bio-fuel, soil amendment, absorbent, industrial uses for the production of biopolymers, carbon black, production of high-performance materials. The hydrochar activated with NaOH at low temperature described by the authors has some important properties: low ash content, high carbon level, low aromatization degree and various oxygenated functional groups. This material exhibited significant adsorption capacity in a basic environment.	S
Maderuelo-Sanz et al., 2018	Sound absorber, a porous or fibrous material such as textiles, fleece, carpets, foams, mineral wool, cotton wool, able to absorb sound energy as they damp the oscillation of the air particles by friction. The material described by the authors is a porous sound absorber obtained from CBs. It is most effective by increasing the bulk density, increasing porosity and reducing flow resistivity.	S
Murugan et al., 2018	Pesticide based on the effectiveness of nicotine and its derivatives. The product described by the authors is a pesticide Ag nanostructures derived from CBs, with highly toxic to malaria vectors and with little impact on predatory copepods.	L
Wadalkar et al., 2018	Precast concrete blocks for paving, a construction product obtained by mixing cement, fine aggregate, coarse aggregate, water, admixtures to which a specific shape is given. This solution is used in areas where conventional types of construction are less durable due to many operational and environmental constraints. Paving done with concrete paver blocks is versatile, aesthetically attractive, functional, cost effective and requires less or no maintenance if correctly manufactured and laid. The product described by the authors is a precast concrete blocks for paving mixed with CBs which, according to the tests performed, increases water absorption and flexural strength, while there are no significant variations in the compressive strength.	S
Xiong et al., 2018a	Hydrophobic absorbent, a material having capacity to adsorb specific substance, but not water. This type of product has several practical applications. The product developed by the authors is a superhydrophobic adsorber fiber obtained from CBs that can be rapidly used to treat water pollution caused by oil-spills. The authors have demonstrated its cost-effective, highly selective and environmentally-friendly properties.	S
Xiong et al., 2018b	Supercapacitors is a high-capacity condenser used in applications that require many rapid charge/discharge cycles, rather than compact long-term energy storage. Supercapacitors may be carbon material-based or metal oxide and a conducting polymer-based. They are one of the most potential energy devices with numerous applications in different industrial and civil sectors. The product developed by the authors is a nitrogen-doped hierarchical porous carbons by a combination of UCF and polypyrrole (PPy), characterized by a very high surface area with micro-, meso- and macroporous structures that, thanks to its large specific capacitance and good retention, make it as a potential candidate of a carbon precursor for high-performance supercapacitors.	S
Blankenship and Mokaya, 2017	Hydrogen stores materials span many approaches thanks to the use of materials with a strong interaction with hydrogen or without any reaction. These include: high pressures, cryogenics, and chemical compounds that reversibly release H2 upon heating. The porous carbons material obtained from the authors through the CBs treatment has been shown to be super-porous, to have a ultra-high surface area and pore volume and an oxygen-rich nature, making it particularly effective in hydrogen storage capacity. This material, compared to the others available, have new or improved properties, are easier and cheaper to prepare and are sustainable.	S
Ghosh et al., 2017	Electrically conductive device that have the capability to supply high levels of power and energy. The authors, using CBs, have obtained a greyish white powdery pyrolyzed product, a conducting high-power material with eco-sustainable properties and with applications such as electrical vehicles, micro sensors and portable electronics.	S
Gómez Escobar and Maderuelo Sanz, 2017	Sound porous absorber, material such as textiles, fleece, carpets, foams, mineral wool, cotton wool able to absorb sound energy as they damp the oscillation of the air particles by friction. The authors developed and analyzed a product deriving from the treatment of CBs that demonstrated high sound absorption performance, even better than some commercial absorbers.	S

(*continued on next page*)

Table 4 (*continued*)

	Type of results	Physical state (S = solid; L = liquid; G = gas)
Hamzah and Umar, 2017	Active carbon, which is a form of carbon that has been processed to have small and low-volume pores, increasing the area available for adsorption or chemical reactions. It is valuable for a variety of applications, especially to oil-water separation, methylene blue (MB) adsorption or as super capacitors. The authors developed an activated carbon via KOH chemical activation using microwave and identified its good capacity for absorbing MB.	S
Mohajerani et al., 2017	Asphalt concrete, a construction product obtained by mixing cement, fine aggregate, coarse aggregate, water and admixtures. The authors describe a product obtained encapsulating CBs with bitumen and paraffin wax and incorporating them into asphalt concrete for pavement construction, demonstrating good mechanical and volumetric properties including stability, flow, resilient modulus, bulk density, maximum density and air voids.	S
Ogundare et al., 2017	Nanocellulose. It is a nano-structured cellulose that can be of different types (in relation to their morphology and source): nanocrystal cellulose, cellulose nanofibers or bacterial nanocellulose. The mechanical and chemical properties of the nanocellulose make it extremely suitable for different types of applications. The authors describe the isolation of nanocrystalline cellulose from CA with important properties, such as: biocompatibility, biodegradation, high strength, specific surfacearea, high crystallinity index, low toxicity and density.	S
Suárez-Rodríguez and Garcia, 2017	CBs, as they are available. The authors analyzed the contribution of CBs in response to ectoparasites, in particular how they are used by house finches, highlighting how cigarettes are added in response to an increase in the number of ectoparasites in the nest and thus how they can be regarded as a form of self-medication.	S
Teixeira et al., 2017	Cellulose pulp, the fibrous material deriving from the chemical or mechanical separation of cellulose fibers from wood, fiber crops, waste paper or rags. The authors describe the result of the treatment of CBs for pulp cellulose production. After the evaluation of physical and optical parameters of the cellulose for commercial use, the pulp obtained can be used in the paper industry. A dark liquor is generated by the process and contains lignin, carbonyls, metals, nicotine and specific tobacco nitrosamines. This, even after several treatments, continues to keep an high COD (chemical oxygen demand))	S
Mohajerani et al., 2016	Fired clay bricks. A building material used for the construction of masonry construction. Generally, their composition involves the use of clay-bearing soil, sand, and lime or concrete materials. The authors describe and analyze the characteristics of fired clay bricks produced by mixing CBs with the compound. The results were positive, highlighting their potential use.	S
Nasution, 2016	Pesticide used to kill insect pests thanks to the toxic substances contained in tobacco (in particular nicotine). According to the author, CB insecticide can kill both either by touch (spraying directly to the body of insects) and by swallowing (spraying directly to the leaves as the foods of insects). CB insecticide can be called as systemics because it may be absorbed, injected, or fed into the plant or animal to be protected.	L
Kim et al., 2016	Nitrogen (N)-doped carbon with bimodal structure consists of carbons materials with micropores and mesopores that can induce the high surface area, which confers an increased distribution of C–N catalytic center and provide the easier accessibility to active sites for reactants. These forms of carbon materials obtained by tailoring textural and surface chemical properties, have significant electronic and catalytic performances. The authors obtained a bimodal porous nitrogen (N) doped carbon supported Pt composite which, characterized from a physical and electrochemical point of view, demonstrated an higher oxygen reduction reaction activity and durability than those for the commercial carbon. This is important for the industrial development of fuel cells sector.	S
Wang et al., 2016	Hybrid electrode, a particular composition of the electrodes used in superconsenders that allow to improve the performance of traditional electrodes by simultaneously using two charge storage mechanisms, without compromising the structural integrity and operational life of supercapacitors. The authors describe a carbon fiber and vanadium nitride material obtained from recycling, at the same time, of metal ions and CBS which can be used as a precursor to fabricate the active material for supercapacitor electrode. The evaluations conducted showed its good electro-chemical behaviors. This product meets the needs of the market for energy storage devices.	S
Kadir and Mohajerani, 2015	Fired clay bricks. A building material used for the construction of masonry construction. Generally, their composition involves the use of clay-bearing soil, sand, and lime or concrete materials. The authors describe the production of fired clay bricks by incorporating CBs into the mixture, in particular analyzing the impact of compressive and tensile strength on water absorption properties, as well as the effect on gas emissions. According to the authors, CB bricks and high heating rates should be considered as an option or an alternative in the brick industry to minimize the manufacturing cost and time.	S
Lee et al., 2014	Nitrogen-doped porous carbon material. It is a carbons material with micropores and mesopores, with a high surface area and proper pore size distribution are essential in utilizing a large amount of electrolyte ions with quick transfer mobility at the surface of the electrode material. A well-developed meso-/micropore structure and with pseudocapacitive assistance can largely improve supercapacitive performance. The authors describe the product deriving from the treatment of CBs for the production of nitrogen doped (N-doped) meso-/microporous hybrid carbon material that exhibit a satisfactory rate capability and higher specific capacitance with the desired supercapacitive performance. This is a promising materials for use in supercapacitors due to its low cost, high porosity, electronic conductivity and stability.	S
Mondal et al., 2014	Pesticide based on the effectiveness of nicotine and its derivatives. The product described by the authors is an extract derived from the treatment of CBs, with highly toxic to mosquito larvae and that may directly or indirectly affect on mosquito larvicidal activity.	L

(*continued on next page*)

Table 4 (*continued*)

	Type of results	Physical state (S = solid; L = liquid; G = gas)
Soltani et al., 2014	Mesoporous char, a material containing pores having a large internal surface area and adsorbent capacity. It, tipically, can adsorb a wide range of substances including dyes and heavy metals. The product described by the authors is a mesoporous char derived from the treatment of CBs and chemically modified with nitric acid to enhance its surface acidity. Experimental tests showed that this product has an increase in the surface and a considerable reduction in total ash content.	S
Vahidhabanu et al., 2014	Crude extracts, a substance that derives by extracting a part of a raw material, often by using a solvent such as ethanol or water. The authors describe the product deriving from the treatment of CBs which, thanks to the presence of specific chemicals, allows to produce crude extracts that have demonstrated good performance as corrosion inhibitor for J55 oil well tubular steel.	L
Dieng et al., 2013	Crude extracts, a substance that derives by extracting a part of a raw material, often by using a solvent such as ethanol or water. The authors describe the product deriving from the treatment of CBs which, thanks to the presence of specific toxic chemicals, allows to produce a compound that has shown insecticidal properties against mosquito larvae.	L
Sarani and Kadir, 2013	Fired clay bricks. A building material used for the construction of masonry construction. Generally, their composition involves the use of clay-bearing soil, sand, and lime or concrete materials. The authors describe the product resulting from the mixing of CBs in the compound for the production of fired clay bricks and analyze the effects related to its thermal conductivity. The results show that with higher percentages of CBs, the thermal conductivity properties were also improved. Moreover, at the same time, the porosity is also increased thus the density was significantly reduced to become lightweight brick.	S
Suárez-Rodríguez et al., 2013	CBs, as they are available. The authors analyzed the contribution of CBs in response to ectoparasites, in particular how they are used by house sparrows and house finches. The results showed how CBs are added in response to an increase in the number of ectoparasites in the nest and thus they can be regarded as a form of self-medication.	S
Soltani and Yazdi, 2012	Carbonized porous carbon. This is a form of carbon that has been processed to have small and low-volume pores that increase the surface area available for adsorption or chemical reactions. Activated carbon is valuable for a variety of applications. The authors have developed a carbonized porous carbon through the treatment of CBs with an high Brunauer, Emmet and Teller (BET) surface area.	S
Kadir and Mohajerani, 2011	Fired clay bricks. A building material used for the construction of masonry construction. Generally, their composition involves the use of clay-bearing soil, sand and lime or concrete materials. The authors describe the production of fired clay bricks by incorporating CBs into the mixture, analyzing in particular the compressive strength, flexural strength, density, water absorption and thermal conductivity. According to the authors, CBs can be regarded as a potential addition to the raw materials used in the manufacture of lightweight.	S
Sabzali et al., 2011	Biofilm carrier. These elements make up a biological wastewater treatment plant. The authors describe the use of CBs to obtain biofilm reactors with high surface area, high porosity and roughness. They demonstrated good performance for simultaneous removal of organic matters and nutrients from wastewater.	S
Zhao et al., 2010	Corrosion inhibition extracts, a substance that derives by extracting from a raw material, often using a solvent such as ethanol or water. The authors describe the product deriving from the treatment of CBs which, thanks to the presence of specific chemicals, allows to produce crude extracts that have shown good performance as corrosion inhibitor for N80 Steel.	L
Kadir et al., 2009	Fired clay bricks. A building material used for the construction of masonry construction. Generally, their composition involves the use of clay-bearing soil, sand and lime or concrete materials. The authors describe the production of fired clay bricks by incorporating CBs into the mixture, analyzing in particular the density, strength, thermal conductivity and leachate characteristics.	S
Kadir and Mohajerani, 2008	Fired clay bricks. A building material used for the construction of masonry construction. Generally, their composition involves the use of clay-bearing soil, sand and lime or concrete materials. The authors describe the production of fired clay bricks by incorporating CBs into the mixture, analyzing in particular the compressive strength, flexural strength, density, water absorption and thermal conductivity.	S
Polarz et al., 2002	Porous carbon structures, that is a form of carbon that has been processed to have small and low-volume pores that increase the surface area available for adsorption or chemical reactions. Activated carbon is valuable for a variety of applications. The authors developed a porous carbon material that is nearly completely transformed into nongraphitic carbon and shows a complex pore structure that could find applications in the field of gas purification or more generally as filters.	S

resistant mosquito vectors. The authors used mimicked smoked CBs. To obtain these samples, they applied a simulation approach using a disposable syringe. The simulation was conducted by channelling the burning tip of a cigarette attached to the hub to the barrel by pulling up the plunger. A typical smoker takes 10 puffs of a single cigarette. The results show the Aedes aegypti vulnerability to Marlboro butts during its development and suggest their vulnerability to CBs. However this effect was seen most during the early developmental phases and in the presence of increased amounts of cigarette remnants.

Teixeira et al. (2017) used CBs and waste from filter production in the cigarette industry for cellulose pulp production. Using their own treatment technology, they obtained a cellulose pulp and a viscous dark liquor. The pulp can be used in the paper industry, while the liquid requires treatment since it contains a highly resistant organic content.

All the other studies focus only on the use of smoked filters and present the results of their research without comparing them with clean samples.

4.2. Methodology/treatment system

The second criterion used to analyse the selected papers was the sample processing applied. The approaches were classified into physical, chemical or physical-chemical systems. The results are shown in Table 3.

Table 5
Description of possible uses.

	Possible use	Sector
Anmei et al., 2018	Identification of Sudan I in foods	Chemical analysis
Bandi et al., 2018	Fluorescence response of CBCDs to various biologically and environmentally important metal ions, such as Fe2+, Ca2+, Na+, Pb2+, Cu2+, Mn2+, Cd2+, Sn2+, Cr3+, Al3+, Ni2+, Hg2+, Mg2+, Ba2+, K+, Zn2+, and Fe3	Chemical analysis
Cui et al., 2018	Asphalt for streets	Infrastructure
Ifelebuegu et al., 2018	Adsorbent cellulose for application in oil/water separation and spill clean-up	Environmental engineering
Lima et al., 2018	Hydrochars adsorbent for the removal of methylene blue (MB) from aqueous solutions	Environmental engineering
Maderuelo-Sanz et al., 2018	Cellulose acetate as an insulator	Buildings and structures
Murugan et al., 2018	Highly effective antiplasmodials and insecticides	Insecticide
Wadalkar et al., 2018	Prefabricated concrete paving	Infrastructure
Xiong et al., 2018a	Cellulose as a superhydrophobic absorbent for oil spillage clean-up	Environmental engineering
Xiong et al., 2018b	Potential candidate of a carbon precursor for high-performance supercapacitors	Energy storage devices
Blankenship and Mokaya, 2017	High surface porous carbon and low volume with high hydrogen storage capacity	Energy storage devices
Ghosh et al., 2017	Good conductive material which can be further used in different electronics field	Energy storage devices
Gómez Escobar and Maderuelo Sanz, 2017	Cellulose acetate as a part of an insulator system	Buildings and structures
Hamzah and Umar, 2017	Active carbon for oil-water separation, methylene blue (MB) adsorption, and super capacitors	Environmental engineering
Mohajerani et al., 2017	Prefabricated concrete paving	Infrastructure
Ogundare et al., 2017	Nanocrystalline cellulose (NCC) with numerous applications including heterogeneous catalysis support in synthesis of organic compounds as well as composites for biomedical purposes.	Chemical and Medical industries
Suárez-Rodríguez and Garcia, 2017	Anti-ectoparasite product	Insecticide
Teixeira et al., 2017	Cellulose pulp	Paper industry
Mohajerani et al., 2016	Fired clay bricks	Buildings and structures
Nasution, 2016	Insecticide	Insecticide
Kim et al., 2016	N-doped carbon (NCF)	Energy storage devices
Wang et al., 2016	Hybrid electrode carbon fiber and vanadium nitride material (CF@VN)	Energy storage devices
Kadir and Mohajerani, 2015	Fired clay bricks	Buildings and structures
Lee et al., 2014	Supercapacitor electrode (CNF)	Energy storage devices
Mondal et al., 2014	Pesticides against malaria vectors	Insecticide
Soltani et al., 2014	Mesoporous char	Chemical analysis
Vahidhabanu et al., 2014	Extract as corrosive inhibiting agent	Metallurgical industry
Dieng et al., 2013	Control tool against an insecticide-resistant mosquito vector	Insecticide
Sarani and Kadir, 2013	Fired clay bricks	Buildings and structures
Suárez-Rodríguez et al., 2013	Anti-ectoparasite product	Insecticide
Soltani and Yazdi, 2012	Carbonized porous carbon	Energy storage devices
Kadir and Mohajerani, 2011	Fired clay bricks	Buildings and structures
Sabzali et al., 2011	Biofilm Carrier for waste water treatment	Environmental engineering
Zhao et al., 2010	Extract as corrosive inhibiting agent	Metallurgical industry
Kadir et al., 2009	Fired clay bricks	Buildings and structures
Kadir and Mohajerani, 2008	Fired clay bricks	Buildings and structures
Polarz et al., 2002	Porous carbon structure	Energy storage devices

The types of treatment of CBs described by the authors were divided equally between the different treatment categories identified. The physical treatments, which were used in 27% of the analysed papers, typically involve separating the filters from the remaining non-smoked tobacco, separating them from the outer paper, crushing and, in some cases, mixing them with other products for the preparation of other materials. The latter condition is well described by Mohajerani et al. (2016); Kadir and Mohajerani (2015); Sarani and Kadir (2013); Kadir and Mohajerani (2011) and Kadir and Mohajerani (2008).

A total of 33% of authors instead adopted only chemical treatment approaches in order to use and enhance CBs. Simple soaking or hydrothermal approaches and the use of chemical solvents are the typical treatment approaches. A total of 32% of the papers used chemical and physical coupled treatments. Typically, the sample undergoes a preliminary physical treatment of cleaning and fragmentation and then chemical treatment in order to obtain the final product. Finally, 8% of the authors did not use any treatment for the CBs which were used in their original form.

Gómez Escobar and Maderuelo Sanz (2017) analysed the samples without any treatment and tested the acoustic absorption performance of smoked and unused CBs. The promising results obtained were compared with a subsequent study (Maderuelo-Sanz et al., 2018), where the CBs underwent physical treatment.

The other two studies that analysed the use of CBs without any kind of preliminary treatment analysed their possible reduction of the ectoparasite load in urban birds. Suárez-Rodríguez and Garcia (2017) and Suárez-Rodríguez et al. (2013) studied how smoked cigarette butts may serve as a parasite repellent in urban bird nests, with indirect consequences also for the transmission via the air of possible negative health effects to other organisms.

4.3. Type of product obtained from the process

Table 4 shows the outcomes of the selected papers according to type of treatment results. For each author the results of the recycling process are identified and classified into solid, liquid or gaseous products.

In most cases (84%) the results obtained after the treatment and recycling of CBs are represented by solid state products. Usually, these are porous carbons with different physico-chemical structures and characteristics which depend on the type of treatment undergone by the sample and the substances and/or technologies used. When the results are composite materials, where the CBs (or part of them) are mixed with other products, in most cases the final result is fired clay bricks or a bitumen mixture.

In 16% of the papers, the results of the treatment process are liquid compounds, successfully tested by the authors as pesticides in agriculture, against particular species of insect diseases or as a corrosion inhibitor thanks to the acidification properties of metallic materials.

Liquid outputs used as an insecticide are described in the study by Murugan et al. (2018) where, an Ag toxic nanostructure was extracted from the CBs and tested, with very promising effects, as an insecticide against the malaria vector Anopheles stephensi, the chloroquine (CQ)-

Table 6
Geographical distribution.

	Country
Anmei et al., 2018	China
Bandi et al., 2018	India
Cui et al., 2018	China
Ifelebuegu et al., 2018	UK
Lima et al., 2018	Brazil
Maderuelo-Sanz et al., 2018	Spain
Murugan et al., 2018	India
Wadalkar et al., 2018	India
Xiong et al., 2018a	China
Xiong et al., 2018b	China
Blankenship and Mokaya, 2017	UK
Ghosh et al., 2017	India
Gómez Escobar and Maderuelo Sanz, 2017	Spain
Hamzah and Umar, 2017	Indonesia
Mohajerani et al., 2017	Australia
Ogundare et al., 2017	South Africa
Suárez-Rodríguez and Garcia, 2017	Mexico
Teixeira et al., 2017	Brazil
Mohajerani et al., 2016	Australia
Nasution, 2016	Indonesia
Kim et al., 2016	Korea
Wang et al., 2016	China
Kadir and Mohajerani, 2015	Australia
Lee et al., 2014	Korea
Mondal et al., 2014	India
Soltani et al., 2014	UK
Vahidhabanu et al., 2014	India
Dieng et al., 2013	Malaysia
Suárez-Rodríguez et al., 2013	Mexico
Soltani and Yazdi, 2012	Malaysia
Suárez-Rodríguez et al., 2013	Australia
Kadir and Mohajerani, 2011	Australia
Sabzali et al., 2011	Iran
Zhao et al., 2010	China
Kadir et al., 2009	Australia
Kadir and Mohajerani, 2008	Australia
Polarz et al., 2002	Canada

resistant malaria parasites, Plasmodium falciparum, and microbial pathogens. Dieng et al. (2013) demonstrated good results as an insecticide against Aedes aegypti larvae, experimenting with liquid solutions at different concentrations. Similarly, Mondal et al. (2014) tested a leachate from CBs against the larvicidal activity of mosquitoes. Although mosquito larvae have developed resistance to water with CBs, the presence of phenolic and primary amine containing functional groups in the percolate demonstrated that the mosquito mortality was absolutely concentration dependent.

Finally, Nasution (2016) also analysed the possibility of using CBs as a precursor to pesticide.

The use of CB leachate as a corrosion inhibitor of metal products and structures was analysed and described by Zhao et al. (2010) who tested it for N80 steel at 90 °C in hydrochloric acid. The results show that inhibition efficiencies of 94.6% and 91.7% in 10% and 15% (wt %) HCl solution were achieved, respectively, by adding 5% (wt %) inhibitor. In 20% HCl solution, a maximum inhibition efficiency of 88.4% was shown by adding 10% inhibitor. Finally, Vahidhabanu et al. (2014) tested this compost for the protection of oil-well tubular steel used in acidification of oil and gas production. The highest inhibition efficiencies of 99% and 61% were obtained for 30 °C and 105 °C, respectively at 6% concentration.

No treatments have led to gases being produced.

4.4. Possible use of treatment output

The analysis results of the possible outputs obtained are summarized in Table 5. The possible applications of the treatment results are classified into eight categories:

1 Chemical analysis
2 Infrastructure/ buildings and structures
3 Energy storage devices
4 Environmental engineering
5 Chemical and medical industries
6 Insecticides

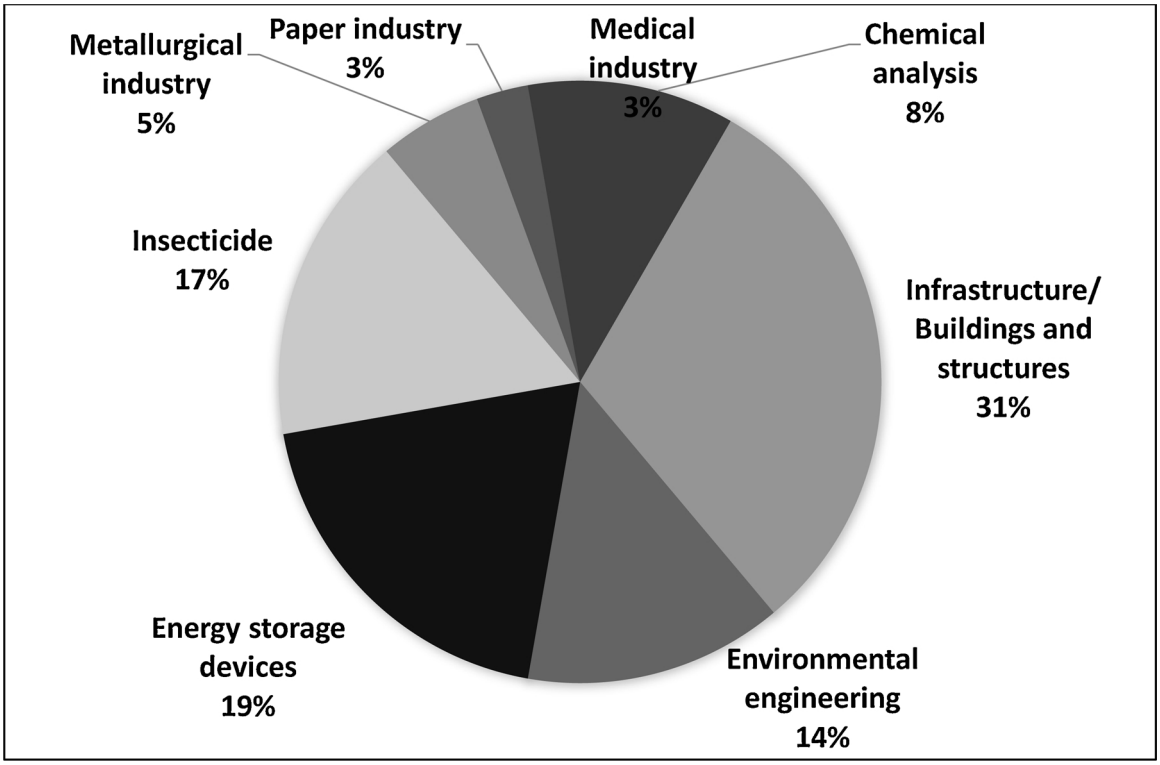


Fig. 8. Percentage distribution of possible uses.

7 Metallurgical industry
8 Paper industry

The results are summarized in Fig. 8, which reports the distribution percentages of the studies among the eight categories.

The solution of using by-products resulting from the treatment processes of CBs in the construction industry and infrastructure has the greatest support among the authors analysed, with 31% of studies describing this possible use.

This research area has been extensively studied by Mohajerani et al. (2016); Kadir and Mohajerani (2015); Sarani and Kadir (2013); Kadir and Mohajerani (2011); Kadir et al. (2009) and Kadir and Mohajerani (2008) and reviewed by Bories et al. (2014).

They evaluated the use of CBs as a mixed component in the preparation of fired bricks, which is considered as common solution to pollution and is already applied with other waste (e.g. rubber, limestone dust and wood sawdust, processed waste tea, fly ash, polystyrene, and sludge).

These authors demonstrated that adding different concentrations of CBs to the compound (from 0% to 10% by weight and corresponding to about 0–30% by volume) for the production of fired bricks, leads to a reduction in the density of the fired bricks, an increase in porosity, and a decrease in thermal conductivity. Low-density or light-weight bricks have several advantages over standard bricks (lower structural dead load, easier handling, lower transport costs, different thermal conductivity), except when stronger bricks are needed. However, this method also leads to a reduction in compressive strength (although in the case studies, the results were lower than the acceptable regulatory Australian limits) which is also important when considering their use. With controlled temperatures, it is also possible to reduce the energy usage during firing and gas emissions into the atmosphere. The related atmospheric emissions have been studied in detail through experimental tests that compared the Estimated Total Emission (ETE) of different types of bricks produced at different heating rates during firing (Mohajerani et al., 2016 and Kadir and Mohajerani, 2015).

These results demonstrate how CBs can be considered as a potential addition to raw materials used in the manufacturing of light fired bricks.

Another use analyzed by the authors is related to the production of asphalt for streets and as prefabricated concrete paving. Mohajerani et al. (2017) studied the possibility of incorporating CBs into asphalt concrete for pavement construction, encapsulating CBs with different classes of bitumen and paraffin wax. Encapsulation prevents chemical translocation from CBs to the environment. Comparing the results obtained in the two cases with the control asphalt concrete sample (no CBs), using mechanical and volumetric properties, including stability, flow, resilient modulus, bulk density, maximum density, air voids, and voids in mineral aggregates as variables, the proposed solutions were found to satisfy the requirements for light, medium and heavy traffic conditions for bitumen samplers, while paraffin wax only satisfied the light traffic conditions for road pavements.

The authors identified this solution as a useful contribution to the decrease in urban heat effect in urban environments due to the reduction in bulk density, the increase in porosity and the reduction of thermal conductivity

A similar study was also carried out by Cui et al. (2018) and Wadalkar et al. (2018). They analyzed the possible benefits of incorporating CBs into precast concrete blocks for paving, assessing their performance as the percentage of added CB increases. They observed that the compressive strength decreased with an increase in the proportion of CBs, while flexural strength and water absorption increased.

To exploit the sound insulation capabilities of CBs, Maderuelo-Sanz et al. (2018) and Gómez Escobar and Maderuelo Sanz (2017) demonstrated the high sound absorption performance of this material and the potential use of CBs as a component for insulating solutions, which were found to be highly competitive even with commercial solutions.

In the field of environmental engineering, the products deriving from the recycling of CBs are above all applied in the treatment of water for oil / water separation, exploiting the absorbing power of CA obtained from CBs. Similar applications have been described by several authors (14% of the total analyzed). Often, the proposed solutions are applicable to accidental conditions, with the release of oil, methylene blue (MB) or liquid petroleum derivatives. Among the various solutions (natural, chemical, and mechanical remediation), the use of oil-absorbent materials is one of the most economical and effective approaches, due to the low operational costs, easy operation and high efficiency. The tests performed have shown very promising results:

- 25.6, 15.8, 17.2, and 21.5 g/g for vegetable oil, diesel oil, motor oil, and vacuum pump (VP) oil, respectively and the sorbent oil sorption capacity did not significantly deteriorate after 20 cycles of reuse (Ifelebuegu et al., 2018).
- maximum monolayer adsorption capacities of 0.3, 0.4, 0.6 and 0.5 g/g (Lima et al., 2018).
- adsorption capacities between 30.6 g/g and 38.3 g/g and high adsorption capacity after 10 cycles (Xiong et al., 2018a).
- adsorption capacities for (MB) of 88.76 m2/g (Hamzah and Umar, 2017).

The use of CBs for energy purposes has also proved to be of interest in the literature. A total of 19% of the authors analyzed reported research on energy, particularly in the field of new, clean, and high-efficiency energy storage devices.

Xiong et al. (2018b) studied the use of CBs as a precursor for the development of supercapacitors, which are one of the best potential energy storage systems thanks to higher energy and power densities than conventional capacitors. They developed a new application of CBs for energy storage, producing carbon with a large specific capacitance and good capacitance retention through the combination of CBs and polypyrrole (PPy). PPy is a conducting polymer used as an electrode material for supercapacitors. The excellent storage performance obtained by the authors was due to the synergistic effects of the porous backbone structure derived from CBs and the pseudocapacitance behavior of PPy-derived nitrogen functionality.

Lee et al. (2014) also developed a supercapacitor based on CBs as the active material, producing a nitrogen doped (N-doped) meso-/microporous hybrid carbon material (NCF), which is a porous carbon material that spontaneously contains both mesopores and micropores. The results highlighted a good capability and higher specific capacitance during 6000 charge and discharge cycles.

In relation to hydrogen, as an attractive energy source especially for motor vehicles, Blankenship and Mokaya (2017) demonstrated that porous carbons derived from CBs with a super porous, ultra-high surface area and pore volume can reach for storing hydrogen. The combination of high surface area, high microporosity and the oxygen-rich nature of the carbons material, ensures a storage capacity of 8.1 wt% excess uptake, and 9.4 wt% total uptake at −196 °C and 20 bar, rising to a total uptake of 10.4 wt% and 11.2 wt% at 30 and 40 bar, respectively.

Kim et al. (2016) developed highly efficient catalysts for the oxygen reduction reaction based on CBs. This is a bimodal porous nitrogen (N) doped carbon supported Pt composite for the industrial development of fuel cells.

Ghosh et al. (2017) showed how pyrolysis treatment produces a material with good conductive properties which can be further used in various electronic fields.

Wang et al. (2016) used waste metal ions and CBs at the same time to develop a hybrid electrode material of carbon fiber and vanadium nitride (CF@VN) with a hierarchical porous structure. This was used as a negative electrode to develop an asymmetric supercapacitor, using an activated carbon as a positive electrode. The resulting product showed good electro-chemical behavior.

Finally, the use of CBs as precursors of carbonized porous carbon has been studied by Soltani and Yazdi (2012) and by Polarz et al. (2002), demonstrating the physical properties that make this material suitable for different purposes, also in electrical and energy storage devices. In the industrial sector, possible applications have been described in the fields of chemical analysis (8%), metallurgy (5%), the paper industry (3%) and the medical industry (3%).

Anmei et al. (2018) describe the use of CBs as precursors for the development of carbon quantum dots (nanomaterial with sizes below 10 nm) which, thanks to their strong fluorescence, have been tested as an agent for the detection of Sudan I in foods. Chilli powder, chilli sauce and tomato sauce were selected as test samples. A similar product is described by Bandi et al. (2018), who used the fluorescence of their N, S-Coded carbon dots for detecting various biologically and environmentally important metal ions, such as Fe2+, Ca2+, Na+, Pb2+, Cu2+, Mn2+, Cd2+, Sn2+, Cr3+, Al3+, Ni2+, Hg2+, Mg2+, Ba2+, K+, Zn2+, and Fe3.

The extract of CBs has also been applied in the metallurgical industry. As described by Vahidhabanu et al. (2014) and Zhao et al. (2010), the resulting solution can be used as a corrosive inhibiting agent in steel materials. Vahidhabanu et al. (2014) used it for the J55 oil well tubular steel used in acidization of oil wells and gas production with efficiencies of 99% and 61% obtained for 30 °C and 105 °C respectively at 6% concentration. Zhao et al. (2010) used the crude extracts from littered cigarette butts for N80 steel in HCl solutions at 90 °C. The inhibition efficiencies were shown to be 94.6% and 91.7% in 10% and 15% (wt %) HCl solution, respectively, by adding 5% (wt %) inhibitor.

In 20% HCl solution, a maximum inhibition efficiency of 88.4% was shown by adding 10% inhibitor.

Teixeira et al. (2017) described the treatment of CBs in order to provide secondary raw material for the paper industry. In fact, CBs are recycled into a cellulose pulp that can be used in the paper industry, either pure or blended with cellulose pulp from various origins depending on availability or the final desired properties.

A possible use in the medical industry is described by Ogundare et al. (2017), who studied the production of nanocrystalline cellulose (NCC) with numerous applications, including composites for biomedical purposes.

Finally, a review of the studies on the recycling of CBs showed a further application discussed by 17% of authors - as insecticides, pesticides or in response to ectoparasites. This solution has been developed in Asia and Central America to counter the spread of parasites and pathogens, especially those transmitted by mosquitoes or birds. These applications are the result of using CBs as they are, or through a liquid compound made from their percolation.

Murugan et al. (2018); Suárez-Rodríguez and Garcia (2017); Nasution (2016); Mondal et al. (2014); Dieng et al. (2013) and Suárez-Rodríguez et al. (2013) analyzed various combinations and concentrations of these solutions, analyzing their effects on the main reference targets.

4.5. Geographical distribution

The geographical distribution of the studies (Fig. 9) shows a strong concentration in Asia, Central and South America and in Australia. Europe is represented by Spain and the UK. Few authors are from North America or Africa.

China, which is the world's leading cigarette consumer with growing values in recent years (2043 cigarettes smoked per person per year), has conducted numerous studies on exploiting CB treatments. Australia has focused on their use for the production of fired clay bricks.

In India, where 89 cigarettes are smoked per person per year (i.e. Indians fall within the average consumption around the world) there have been six studies, two of which are on the development of insecticides to control the spread of diseases through insects.

5. Conclusions and prospects for development

Despite cigarette butts (CBs) being one of the most common types of litter worldwide, with a production of one million tonnes each year, the solution to the problem still appears very challenging and distant.

In fact, this involves at least two main aspects:

1 Consumers: increasing awareness of the environmental impact of CBs and changing their habits in dealing with this waste (today in most cases still discarded on the ground). Policies and awareness campaigns are used to achieve this.
2 Management methodologies, identifying one (or more) economical and effective technological solutions for the prevention or treatment of this waste and also identifying the best approaches for the collection and transport of CBs.

This paper has analysed the second aspect through a general review of 37 selected papers that provide different types of solutions for the treatment and recycling of CBs.

The following conclusions can be drawn from the results presented in this paper:

- CBs constitute widespread waste globally and are often more concentrated in areas with the highest number of consumers. China is an example of this situation: people smoke over 40% of all cigarettes globally (2043 cigarettes per person per year) and has the largest number of CBs and packaging waste in the world.
- This aspect also influences the origin of the studies analysed, which are often concentrated in areas where the problem appears to be very significant or, as in the case of India and Mexico, where local conditions require dedicated solutions (e.g. integrated vector management strategies).
- Comparing the authors' approaches to the treatment of CBs, in most tests only smoked CBs were used, often without providing comparative analyses of the same non-smoked material (CFs cleaned).
- The preparation and treatment methods of the samples are mainly of a physical nature, through which the CBs are separated from the remaining non-smoked tobacco, and from the outer paper and crushed into small fragments.
- The results of the treatment processes constitute in particular solid materials used in numerous economic and productive sectors, especially in the field of building materials, energy and the environment and chemistry.
- Few authors describe the methods of collecting the CB samples. In fact, laboratory applications do not require large quantities of samples. However, the logistical aspect associated with the collection and handling of CBs is an extremely critical element of the process, which risks making the product developed ineffective and the economic system not economically viable.
- The treatment processes described all take place on a laboratory pilot scale. However, a large-scale analysis or experimental application is lacking, which would simulate the operational process with greater quantities and with the critical, operational, economic and legislative barriers that characterize a pilot case.
- With the exception of very few studies, most are very recent scientific contributions. This shows how the correct management of CBs represents a current priority of global interest.
- All the studies analysed describe the methodological and operative treatments of CBs and analyse the results and the products obtained in detail. This provides very detailed information on the characteristics of the resulting materials and performance under operating conditions. However, no author provides detailed information regarding the economic sustainability of the process or how to make these products suitable for the market.
- Finally, the review highlights that social aspects are still poorly investigated and, in most cases, neglected. Including social factors

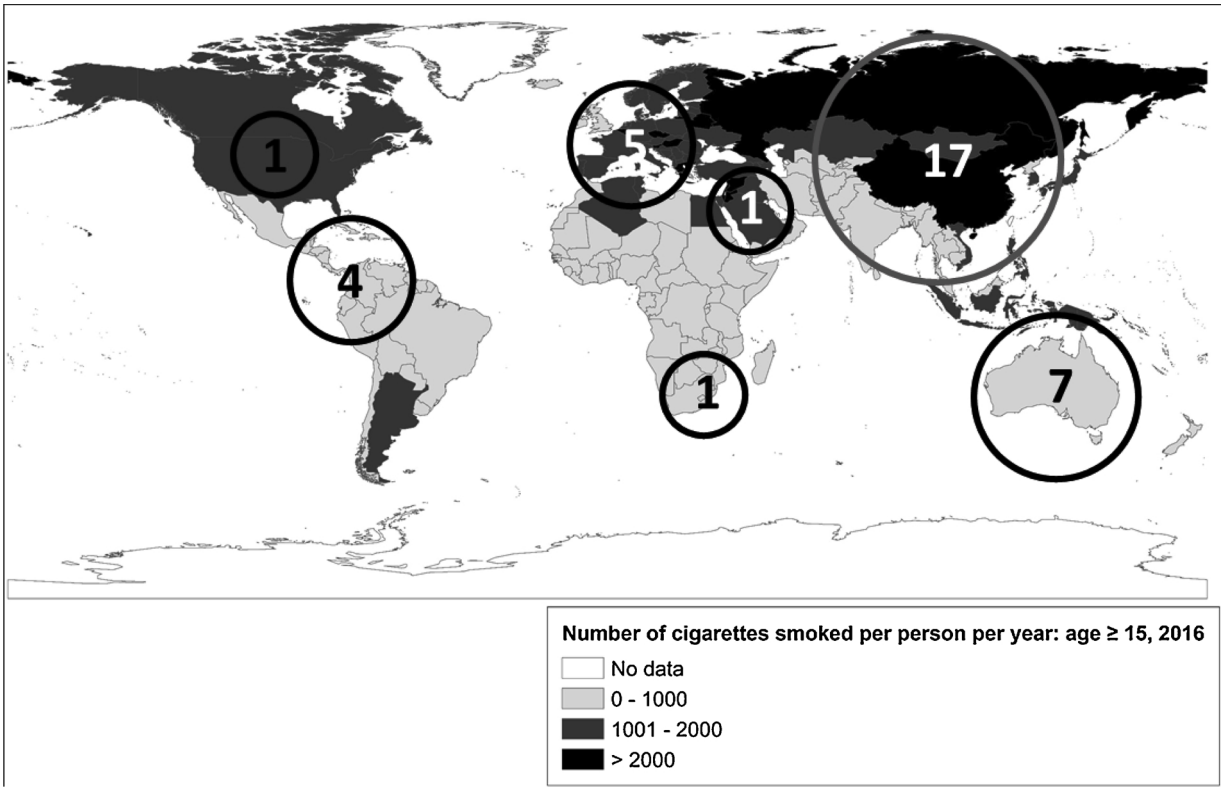


Fig. 9. Spatial distribution of studies and pro-capita cigarette consumption (authors elaboration).

would significantly influence the functioning and effectiveness of the described treatments. One example is the propensity of citizens to discard the CBs in the appropriate bins, thus encouraging the collection of samples at established points.

The economic sustainability of the processes, the consumption of energy and/or reagents or other products, as well as the toxicity of the outputs are not dealt with by most of the authors analyzed. Also, there is no information on the yields of the processes and the possibility of passing from the laboratory scale to the industrial scale. These are critical aspects common to most papers. Table 7 gives a brief critique of each article, highlighting the main advantages and disadvantages of each solution.

Future scenarios could evaluate the transfer and capitalization of the proposed studies and technologies, through an analysis of the following components:

- The supply chain.
- The market.
- The technologies tested through a full-scale pilot.
- The economic sustainability.
- A business and capitalization plan.

Despite the interesting and promising results presented in the literature, CBs remain a particularly critical waste due to their characteristics and impact on the environment and on the health of living organisms. Therefore, the main challenge of preventing the production and spread of this waste remains.

Avoiding the production of waste according to the "precautionary principle" remains the optimal solution that must be increasingly applied and disseminated (WHO, 2017). The application of this principle is not simple, therefore solutions are also needed to avoid dispersion, that would facilitate the correct collection and management of CBs.

Below are some examples of possible measures to reduce the production and dispersion of CBs in the environment:

- Labeling the danger. Just as health risks are reported on packets, this information could also state that CBs and packaging products are a non-biodegradable hazardous waste. A description of the main effects on the environment could be included together with some recommendations on the correct disposal methods to reduce their dispersion.
- Using biodegradable materials and banning disposable filters. The use of biodegradable materials for the production of CFs and packaging products could be encouraged. There are already various international initiatives that reduce the impact and increase the sustainability of the box, inner liner, ink, filter, tobacco and cigarette paper.
- A "Deposit / Return" approach. This approach has already been successfully applied to other materials, for example glass bottles. The same approach, which includes a deposit for the purchase, could also be applied to the sale of cigarette packets. The “butt deposit” could be refunded when the packet is returned to the vender with the butts.
- A waste tax. A specific tax could be introduced that is added to the cost of cigarettes and used to finance specific activities, such as cleaning up local areas, financing research projects, and so on.
- Anti-smoking policies and waste fines. This solution requires political intervention by local administrations that could ban smoking in certain critical areas. The penalty for offenders would be a fine. This solution has already been tested: for example in 2019 several Italian beaches have been smoke-free. This prevents the presence and accumulation of CBs in critical areas, such as coastal areas.
- Litigation against the tobacco industry. This solution requires specific even economic interventions of the tobacco industries to promote solutions to the production and dispersion of CBs in the environment. Often, expensive legal battles are needed to obtain the support of the industry.
- CB collection campaigns. This solution is already internationally widespread with teams of volunteers involved in waste collection (including CBs) in some areas (beaches in particular). Also thanks to

Table 7
Advantages and disadvantages of each approach.

	Advantages	Disadvantages	Pilot or larger scales
Anmei et al., 2018	- Simple and economical hydrothermal process. - High water solubility, photostability, toxicity and biocompatibility.	- Chemical composition of formed CDs is not studied. - The mechanism was not explained.	Pilot scale
Bandi et al., 2018	- Simple preparation method. - Application for the analysis of water (tap, bore, and pond) and biological samples (human urine and serum). - Good production yield. - Cheap.	- No information about energy consumption. - No information about the toxicity.	Pilot scale
Cui et al., 2018	- Very simple process. - CBs can improve pavement performance and save the costs. - - Sustainability assessment was made.	- There is no mixture that allows all the optimal parameters. - Its use is related to environmental conditions.	Pilot scale
Ifelebuegu et al., 2018	- Simple process. - Adsorbs 16 to 26 times their weights in various oils. - Its capacity did not deteriorate after 20 cycles. - Excellent water repellency, - oil sorption and recyclability. - Experimented with different types of oil.	- No information about toxicity. - The treatment requires the use of chemical reagents.	Pilot scale
Lima et al., 2018	- Production at low temperatures. - High absorbent capacity.	- Experimented only with MB. - Significant capacity for adsorption in a basic environment.	Pilot scale
Maderuelo-Sanz et al., 2018	- Absorbent performance similar to commercial products. - Good broadband sound absorbers. - Comparison also between used and not used CA.	- Further studies are needed for the problem of odour and to treat the potentially harmful constituents.	Pilot scale
Murugan et al., 2018	- Low doses of CBs extracts (with and without - tobacco) showed larvicidal and pupicidal toxicity. - No significant effects on non-target risks.	- Other toxicological studies may be useful.	Pilot scale
Wadalkar et al., 2018	- Very simple process. - Recommended grades of paver blocks for different traffic categories were considered.	- Need for preliminary treatment. - Water absorption increased with the increase in percentage of CBs. - There is no mixture that allows all the optimal parameters.	Pilot scale
Xiong et al., 2018a	- Simple method. - Good absorption capacity. - Adsorbed oil was readily and rapidly recovered by mechanical squeezing. - Can be re-used in different cycles.	- No information about toxicity. - Tested on a few cycles. - Requires chemical reagents for its production.	Pilot scale
Xiong et al., 2018b	- Simple method. - Carbon material with high specific surface area, numerous nitrogen atom-containing groups, hierarchical porous structure and good specific capacitance. - Resistant to numerous cycles.	- No information about energy consumption.	Pilot scale
Blankenship and Mokaya, 2017	- Super porous, with ultra-high surface area and pore volume. - The carbons exhibit unprecedentedly high hydrogen storage capacity. - Low cost.	- No information about energy consumption. - No information about the yield.	Pilot scale
Ghosh et al., 2017	- Simple process. - No pre-treatment is required. - Extensive applications in the electronics sector.	- No information on further processing for industrial use.	Pilot scale
Gómez Escobar and Maderuelo Sanz, 2017	- High sound absorption performance. - Comparison with other materials.	- Preliminary results. - Other studies are needed.	Pilot scale
Hamzah and Umar, 2017	- First study about the preparation of active carbon from CBs by microwave induced KOH activation. - Microwave heating method guarantees rapidly and uniformly heats and it is an energy-efficient method.	- No information about energy consumption.	Pilot scale
Mohajerani et al., 2017	- Simple method. - Satisfactory for light, medium and heavy traffic conditions. - CBs increases the porosity which, in turn, lowers its thermal conductivity. - Different bitumen classes were used.	- Content is limited and should not exceed 3%. - Tests were carried out in accordance with (only) the relevant Australian standards.	Pilot scale
Ogundare et al., 2017	- First study about the isolation and characterization of NCC derived directly from DCF. - NCC has numerous applications.	- Requires several steps and the addition of solvents. - The isolated NCC-DCF will be further investigated for such applications.	Pilot scale
Suárez-Rodríguez and Garcia, 2017	- Disassembled CBs may be brought to the nests because they repel ectoparasites.	- Process not under control. - Missing information on the toxic effect.	Pilot scale
Teixeira et al., 2017	- Technology for separating all the cellulose content in the CBs. - The cellulose pulp can be used in the paper industry.	- Process consisting of several steps. - Remains a dark liquor which, despite various treatments, maintains a high COD.	Pilot scale
Mohajerani et al., 2016	- Simple process. - Assessed several% of CBs. - Analyzed different physical characteristics. - Analyzed atmosphere emissions. - Leachate analyses. - Analyzed energy savings.	- Other methods of reducing gas emissions need to be investigated. - Need for tests in a factory environment.	Pilot scale

(*continued on next page*)

Table 7 (*continued*)

	Advantages	Disadvantages	Pilot or larger scales
Nasution, 2016	- Simple process. - It can be used in any situation, anytime and in any place.	- Need more research in the field. - Need self-consciousness. - Less accuracy.	Pilot scale
Kim et al., 2016	- Simple process. - The presence of N dopant on the surface of the NCF not only provided highly dispersive active sites for the growth of the Pt nanoparticles, but also the active centers for ORR itself. - Large surface area and fast oxygen insertion and desertion.	- NCF is a promising supportmaterial, but further investigation is needed - No information on energy consumption.	Pilot scale
Wang et al., 2016	- Simple, inexpensive, and effective approach. - Recycle waste metal ions and CBs at the same time. - -Exhibited high electrochemical performances and cycling stability.	- No information about energy consumption. - No information about the toxicity.	Pilot scale
Kadir and Mohajerani, 2015	- Study the effect of heating rate on gas emissions. - Gas emissions were reduced significantly with higher heating rates.	- No information about energy consumption.	Pilot scale
Lee et al., 2014	- High rate capability and high specific capacitance. - Synergistic combination of well-developed meso-/ - micropores, an enlarged surface area and pseudocapacitive behavior leads to the desired - supercapacitive performance. - Good performance even after many cycles (6.000).	- No information about energy consumption. - Further exploration of CFs for a supercapacitor electrode is warranted.	Pilot scale
Mondal et al., 2014	- Simple process. - The mosquito mortality is concentration dependent.	- Only tested on the mosquito. - More extensive research is needed.	Pilot scale
Soltani et al., 2014	- Considerable reduction in total ash content of the surface-modified char compared to the non-modified char.	- Chemical modifications are needed after the pyrolysis process.	Pilot scale
Vahidhabanu et al., 2014	- Simple process. - Good performance as corrosion inhibitive effects.	- Uses chemical solvent. - There is no toxicological information.	Pilot scale
Dieng et al., 2013	- Simple process. - In great presence of CBs, mortality was high even for the late developmental stages.	- Evaluation limited to Aedes aegypti vulnerability. - Only one type of cigarette was evaluated.	Pilot scale
Sarani and Kadir, 2013	- Compare different % of CBs and different heating rates. - -With higher % of CBs, the thermal conductivity properties were also improved.	- Analyzed only the thermal conductivity.	Pilot scale
Suárez-Rodríguez et al., 2013	- CBs may function as a parasite repellent in urban bird nests. - Assess smoked or non-smoked cigarette butts. - No treatment is necessary.	- No toxicological evidence has been studied.	Pilot scale
Soltani and Yazdi, 2012	- Comparison between impregnated and non-impregnated samples. - High Brunauer, Emmet and Teller (BET) surface area.	- Dry KOH impregnation of cigarette butts as carbon precursor would not be a suitable approach in obtaining higher BET surface area.	Pilot scale
Kadir and Mohajerani, 2011	- Simple process. - Different percentages of CBs were incorporated into manufactured fired clay bricks. - The effects on physical and mechanical properties of samples were investigated. - The effect of the incorporation of CBs on porosity and thermal conductivity of bricks was also investigated and a predictive model was developed.	- Preliminary investigation on recycling CBs in clay bricks. - More extensive experimental work is recommended, using - different types of clay soils. - Further research should also focus on the use of potential cost-effective additives/waste such as fly ash.	Pilot scale
Sabzali et al., 2011	- Comparison of integrated fixed film activated sludge - (IFAS) and activated sludge (AS) system. - Good effect on ammonia and COD removal. - Interesting results on removal efficiency of total phosphorus (TP-P). - High surface area, high porosity and roughness of this carrier can increase the performance. - This process can tolerate strong loading shock.	- Different steps are needed. - No information about energy consumption.	Pilot scale
Zhao et al., 2010	- Good inhibition efficiencies. - Inhibition efficiencies are related to the concentration of CBs water extracts.	- No information about energy consumption. - Just one tested material: N80.	Pilot scale
Kadir et al., 2009	- Simple process. - Different percentages of CBs were incorporated into manufactured fired clay bricks. - The density, compressive strength, water absorption and amounts of heavy metals were investigated. - The effect of the incorporation of CBs on porosity and thermal conductivity of bricks was also investigated and a predictive model was developed.	- No information about energy consumption. - Preliminary investigation on recycling CBs in clay bricks.	Pilot scale
Kadir and Mohajerani, 2008	- Simple process. - Different percentages of CBs were incorporated into manufactured fired clay bricks. - The density, compressive strength, water absorption and amounts of heavy metals were investigated.	- No information about energy consumption. - Preliminary investigation on recycling CBs in clay bricks.	Pilot scale
Polarz et al., 2002	- Simple process. - High-temperature treatment under inert conditions without extensive preparative methods.	- No information about energy consumption.	Pilot scale

the support of digital technologies and social networks, there are now widespread initiatives aimed at the active involvement of citizens who, given the collection of a certain quantity of CBs (e.g. a glass or a bottle), receive a reward, typically of food or drink.

- Awareness and education campaigns aimed at consumers. This solution involves campaigns aimed at empowering consumers, increasing their sensitivity to the impact of CBs on the environment and encouraging their participation in the correct storage of waste.

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